



MARPA

MARPA IFC Naming And Classification Conventions

Progress memo for MARPA review

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Prepared from the current Vectorworks / IFC pipeline planning work, using MARPA source naming as the starting point.

● Purpose

This memo summarizes the proposed naming and classification approach for the MARPA Vectorworks-to-IFC workflow. The goal is to make Vectorworks exports more useful as BIM, cost-estimation, and digital-twin evidence without losing the original MARPA project naming or Vectorworks context.

The key idea is simple:

Keep Vectorworks names exactly as source evidence, then add a MARPA classification layer that maps each object to IFC entities, IFC property sets, quantity takeoff fields, cost accounting fields, and source lineage.

This is not final table design. It is a naming and IFC-mapping discussion document so MARPA can react early.

● What We Are Standardizing

The naming system has four layers:

Layer	What It Preserves	What MARPA Adds
Source file identity	Original file name, path, file hash, export date	Canonical project id, job number, source type, export package id
Vectorworks identity	Native layer, class, record, worksheet, resource, object name	Stable MARPA object id and normalized category
IFC identity	IFC schema version, entity class, type, GlobalId, Name, Tag	MARPA property sets and classification references
Cost/quantity identity	Area, volume, count, length, material, labor/pricing references	Quantity takeoff fields, cost code, estimate group, confidence, evidence source

The current rule is additive: original names stay intact; MARPA names are additional metadata.

● IFC Naming Basics

IFC already has strict conventions that we should respect:

IFC Concept	Convention	MARPA Use
IFC entity	CamelCase with <code>Ifc</code> prefix, such as <code>IfcSlab</code> , <code>IfcWall</code> , <code>IfcGeographicElement</code>	Select the closest standard entity for each landscape/design object
Standard property set	<code>Pset_*</code> , such as <code>Pset_SiteCommon</code> or <code>Pset_VegetationCommon</code>	Use only when the property set is defined by buildingSMART
Quantity set	<code>Qto_*</code> , such as <code>Qto_WallBaseQuantities</code>	Use for derived quantities such as area, length, count, volume, or weight
Classification	<code>IfcClassificationReference</code>	Attach CSI MasterFormat, internal MARPA cost categories, plant categories, and project-specific codes
Custom property set	Must not use the <code>Pset_</code> prefix	Use <code>MARPA_*</code> property sets for MARPA-specific data

The custom MARPA property sets should therefore use names like `MARPA_ObjectIdentity`, not `Pset_MARPA_ObjectIdentity`.

● Proposed MARPA IFC Property Sets

These are the first proposed custom property sets. They are intentionally grouped by business meaning instead of by software tool.

Property Set	Purpose	Example Fields
<code>MARPA_ObjectIdentity</code>	Stable MARPA object naming	<code>marpa_object_id</code> , <code>project_code</code> , <code>job_number</code> , <code>object_family</code> , <code>object_sequence</code> , <code>canonical_name</code>
<code>MARPA_VectorworksBridge</code>	Preserve Vectorworks source context	<code>vwx_layer</code> , <code>vwx_class</code> , <code>vwx_object_name</code> , <code>vwx_record_format</code> , <code>vwx_worksheet</code> , <code>vwx_resource_name</code>
<code>MARPA_SourceLineage</code>	Show where the object came from	<code>source_file</code> , <code>source_hash</code> , <code>export_type</code> , <code>export_timestamp</code> , <code>source_page</code> , <code>source_sheet</code> , <code>extraction_method</code>

Property Set	Purpose	Example Fields
MARPA_GeometryEnrichment	Track created or corrected geometry	geometry_source, xy_source, z_source, extrusion_height, georef_status, geometry_confidence
MARPA_QuantityTakeoff	Attach quantities used for estimation	quantity_basis, count, length, area, volume, unit, measurement_method
MARPA_CostAccounting	Connect design objects to estimating	cost_code, cost_code_system, estimate_group, material_basis, labor_basis, pricing_source
MARPA_LandscapeAsset	Planting and landscape-specific identity	botanical_name, common_name, plant_size, container_size, spacing, asset_role
MARPA_DecisionRecord	Record human or system decisions without blocking execution	decision_type, decision_value, decided_by, decided_at, notes

● Example Object Naming

Recommended MARPA object id pattern:

MARPA- {PROJECT_CODE} - {JOB_NUMBER} - {OBJECT_FAMILY} - {SEQUENCE}

Examples:

Object	MARPA Object ID	IFC Entity Candidate	IFC Name
Concrete paver walk	MARPA-GANNAN-2515-HARDSCAPE-0001	IfcSlab or IfcCovering	Hardscape - Concrete Paver Walk - H001
Retaining wall	MARPA-GANNAN-2515-WALL-0001	IfcWall	Wall - Retaining - W001
Plant instance	MARPA-GANNAN-2515-PLANT-0001	IfcGeographicElement/VEGETATION	Plant - Woolly Thyme - P001
Planting area	MARPA-GANNAN-2515-PLANTING_AREA-0001	IfcGeographicElement or terrain-related geometry	Planting Area - Groundcover - PA001
Existing tree	MARPA-GANNAN-2515-TREE-0001	IfcGeographicElement/VEGETATION	Existing Tree - T001

The exact IFC entity can vary by export version and receiving software. The important rule is that MARPA keeps the object id stable even if the IFC entity mapping changes.

● Landscape IFC Mapping Notes

Landscape BIM is less standardized than building interiors, so MARPA should expect some practical mapping decisions:

Landscape Item	Preferred Direction	Notes
Hardscape surfaces	<code>IfcSlab</code> , <code>IfcCovering</code> , or related built element	Use real area/thickness/elevation when available; enrich missing Z values
Walls and vertical site elements	<code>IfcWall</code> or closest standard built element	Preserve height, length, material, and section evidence
Plants and trees	<code>IfcGeographicElement</code> with vegetation semantics when supported	Use <code>Pset_VegetationCommon</code> where supported, plus <code>MARPA_LandscapeAsset</code>
Terrain and grading	Site/geographic/terrain representation depending on export	Georeference and coordinate origin are critical
Unknown/custom objects	Avoid generic proxy when a better standard entity exists	Use fallback only when the object cannot be responsibly classified

● Quantity Takeoff Direction

Cost estimation depends on quantities. IFC supports quantities through `IfcElementQuantity` and quantity sets. MARPA should separate:

Quantity Type	Example	Preferred Source
Count	number of plants, number of fixtures	Vectorworks object/resource records or worksheets
Area	paver area, planting bed area, turf area	Vector geometry, worksheet, or enriched geometry
Length	wall length, edging length, pipe length	Vector geometry or design worksheet
Volume	wall volume, soil volume, slab volume	3D geometry or enriched 2D-to-3D reconstruction
Unit cost basis	per each, per square foot, per linear foot, per cubic yard	MARPA cost/pricing source

The naming convention should make the measurement basis visible. For example, a hardscape item should not just say “pavers”; it should carry area, unit, material, and the source of the quantity.

● Proposed File And Export Artifact Naming

Original filenames are preserved. Derived export artifacts can use a canonical MARPA name:

```
marpa_{project_code}_{job_number}_{source_system}_{doc_type}_{sheet_or_layer}_{content_date}
_{version}_{artifact_kind}
```

Example:

```
marpa_gannan_2515_vectorworks_design-layer_hardscape-plan_20260704_v01_pdf
marpa_gannan_2515_vectorworks_ifc4x3_full-model_20260704_v01_ifc
marpa_gannan_2515_mineru_sheet-l2_layout-json_20260704_v01_json
```

This lets MARPA keep the original file while giving every derived artifact a predictable identity.

● What MARPA Can Add Or Change

Useful feedback from MARPA would be:

1. Do you already have preferred Vectorworks layer/class naming standards?
2. Do you want job numbers, client names, or internal project codes in object ids?
3. What cost code system should be primary: CSI MasterFormat, MARPA internal categories, or both?
4. Which objects matter most for estimation: hardscapes, walls, planting, irrigation, lighting, grading, furnishings, or all of them?
5. Do you have preferred plant naming rules: botanical name first, common name first, size/container first, or another convention?
6. Which IFC version do outside partners need: IFC2x3, IFC4, IFC4x3, or multiple exports?
7. Are there specific properties MARPA wants attached to every object for estimating or maintenance?

● Near-Term Implementation Path

The next step is not database loading. The next step is a pilot Vectorworks harvest:

1. Open one current project VWX file.
2. Export IFC2x3, IFC4, and IFC4x3 if available.
3. Export worksheets, PDFs, design-layer images, sheet-layer images, and DWG/DXF.
4. Compare which fields appear in each export.
5. Draft the first MARPA mapping file from real exported objects.
6. Attach `MARPA_*` property sets to a small sample of enriched objects.
7. Re-import or inspect the enriched IFC to confirm the properties survive the round trip.

● Sources

- buildingSMART, Industry Foundation Classes overview: <https://www.buildingsmart.org/standards/bsi-standards/industry-foundation-classes/>
- buildingSMART, IFC 4.3.2 documentation: <https://ifc43-docs.standards.buildingsmart.org/>

- buildingSMART, `IfcPropertySet` : <https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/lexical/IfcPropertySet.htm>
- buildingSMART, quantity sets: https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/concepts/Object_Definition/Quantity_Sets/content.html
- buildingSMART, `IfcElementQuantity` : <https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/lexical/IfcElementQuantity.htm>
- buildingSMART, `IfcGeographicElement` : <https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/lexical/IfcGeographicElement.htm>
- buildingSMART, `Pset_VegetationCommon` : https://ifc43-docs.standards.buildingsmart.org/IFC/RELEASE/IFC4x3/HTML/lexical/Pset_VegetationCommon.htm
- MARPA local evidence handoff: `/Volumes/marpa-volume/DATA-PIPELINE-PLAN/NEXT_VECTORWORKS_HARVEST_HANDOFF_20260704.md`